

Newborough Warren

Why the Water Under the Dunes Matters for Wildlife

A condensed summary of a 21-year groundwater study

Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales. Draft, 2026.

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The place

Newborough Warren sits at the southern tip of Anglesey — 1,300 hectares of sand dunes, wetlands and conifer plantation, designated as a Special Area of Conservation. The heart of the site is its dune slacks: shallow, seasonally flooded hollows between the dunes that are among the rarest habitats in Britain. They support nationally scarce plants including Fen Orchid (*Liparis loeselii*), creeping willow and a rich community of wetland invertebrates that exist nowhere else in Wales at this density. All of them depend on the water table sitting within 37 cm of the surface through the summer. That margin — about the length of a ruler — is the difference between a thriving wet slack and an irreversibly degraded dry one. A 700-hectare Corsican pine plantation, planted between 1948 and 1965, intercepts roughly a quarter of incoming rainfall before it can reach the water table. This study monitored 88 dipwells monthly for 21 years (2005–2026), asking a single ecological question: is the water table staying close enough to the surface to keep the dune slack communities viable?

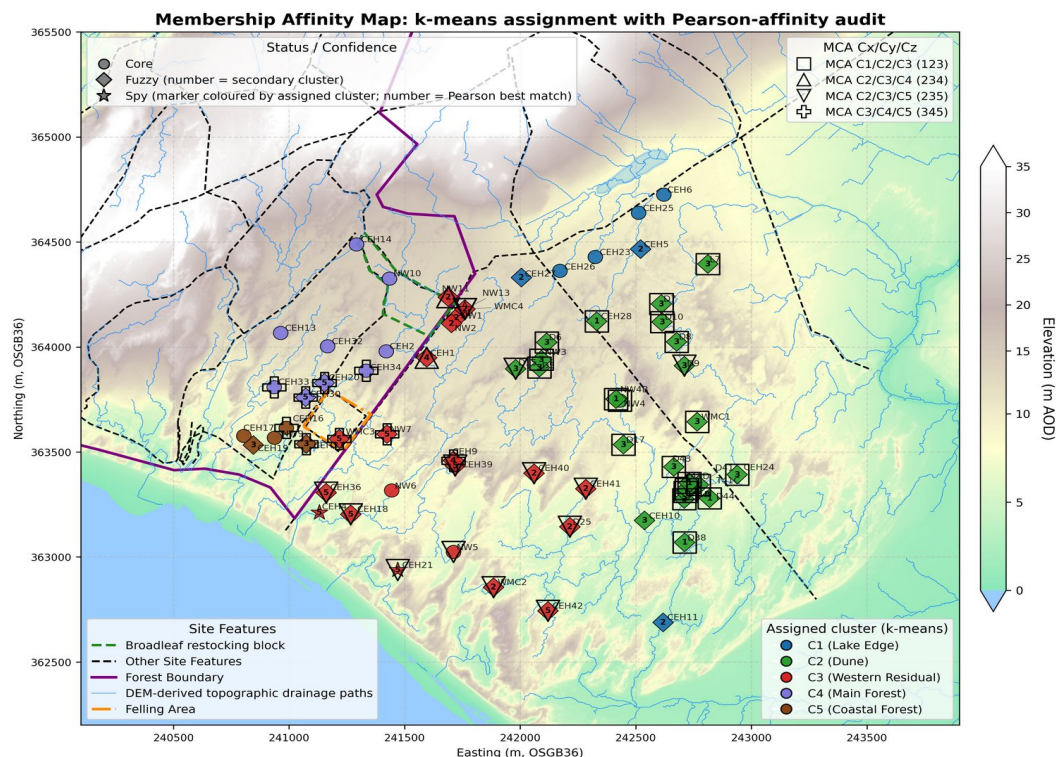


Figure 1. Spatial mapping of the five zones: C1 Lake Edge (blue), C2 Dune (green), C3 Western Residual (red), C4 Main Forest (purple), C5 Coastal Forest (brown).

Five groundwater zones — five ecological situations

A water balance model fitted to each well groups the site into five zones with distinct ecological outlooks. C1 (Lake Edge) is the most immediately threatened zone — shallow, fast-draining, and already recording individual summers where the water table reaches the wet-slack viability limit. C2 (Dune) supports the bulk of the open dune slack habitat; moderate rainfall sensitivity gives it some resilience but long-term decline is measurable. C3 (Western Residual) acts as a slow deep buffer, responding less strongly to individual dry summers but not immune to the long-term trend. C4 (Main Forest) sits on thin sand over irregular bedrock: pine interception and a low-capacity substrate suppress winter recharge; the clearfell is intended to reverse this. C5 (Coastal Forest) is declining fastest of all, driven by coastal retreat as well as pine interception. The key ecological point is that the thickness of sand beneath each area — not just tree cover — determines how severely evaporation draws down the water table in summer.

Zone	Character	Recharge sensitivity	Evaporation draw	Drainage rate	Canopy share
C1 Lake Edge	Shallow, fast-draining	4.58 (highest)	0.96 (lowest)	0.090 (fastest)	0.22
C2 Dune	Open mature dune	3.98	1.77	0.07	0.26
C3 W. Residual	Deep buffer	3.57	1.85	0.06	0.28
C4 Main Forest	Pine on thin bedrock substrate	2.52 (lowest)	2.55 (highest)	0.020 (slowest)	0.4
C5 Coastal Forest	Pine on deeper coastal sand	2.41	1.32	0.05	0.41

Table 1. The five groundwater zones. How much rain reaches the water table (recharge sensitivity), how hard evaporation pulls it down (evaporation draw), how quickly it drains away (drainage rate), and what fraction of the sky is shaded by canopy (canopy share). Higher recharge and lower evaporation draw is better for dune slack ecology.

What the data show

Summer is the critical season

The water table has a critical season and a critical depth. Summer is when plants and invertebrates experience maximum stress: if the water table drops below 61 cm in summer (the Curreli et al. 2013 wet-slack threshold), hygrophilous species cannot survive. If it drops below 98 cm, even dry-slack communities become marginal. The summer minimum has been declining across all five zones. The Lake Edge zone — the zone most directly linked to the orchid populations — is on course to cross the wet-slack threshold as a long-term mean around 2030–2032; individual dry summers already reach it. This is not a distant projection: the window for effective intervention is the next decade. UKCP18 projections indicate summer minima will deepen by a further 71–134 mm by the 2080s. The five-year mean spring baseline (MSL5), used to track management progress, is more stable — projected to deepen by 21–39 mm over the same period, three to five times less than the summer minimum. This divergence reflects the ecological asymmetry of the aquifer: winter recharge is partially buffered by drainage storage, but summer draw-down is not.

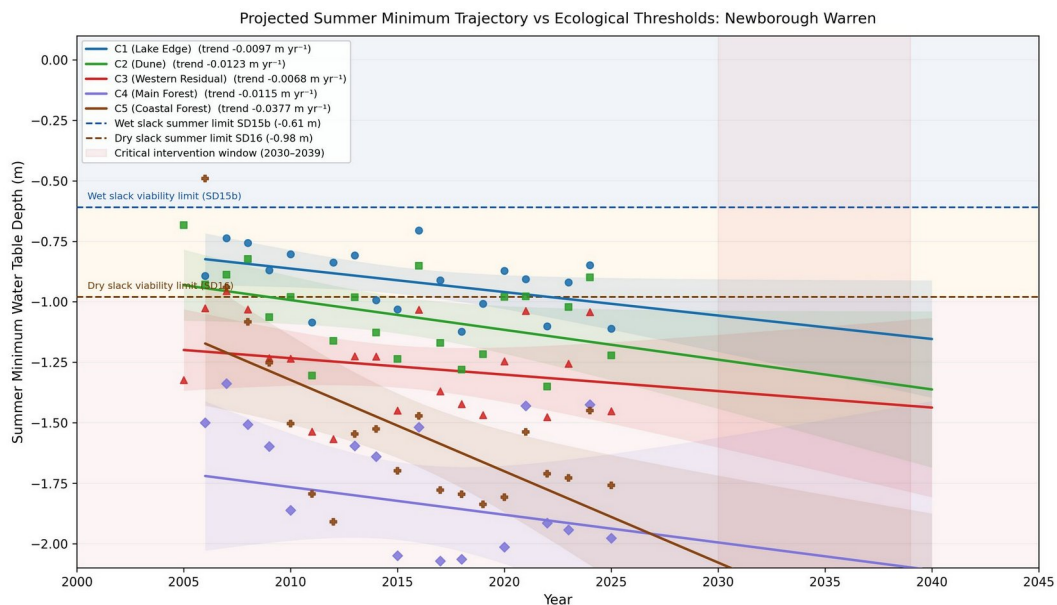


Figure 2. Projected summer minimum depth for all five zones. Wet-slack threshold -0.61 m (blue dashed); dry-slack threshold -0.98 m (brown dashed). Critical intervention window 2030–2039 shaded.

Dune scraping

Scraping removes several centimetres of sand from the slack floor, permanently lowering the ground surface and bringing it closer to the water table. The ecological effect is immediate and geometric: a slack scraped 20 cm deeper sits 20 cm closer to the water table for as long as the scrape geometry is maintained. At CEH36 (scraped April 2015), three independent analytical methods converge on a scraping benefit of 80–140 mm in mean water-table depth; the ecologically relevant test — the shift in summer minimum depth against the unscraped control well — was +195 mm ($p = 0.004$). That shift moved CEH36 from a zone where the wet-slack threshold was routinely exceeded to one where it is only reached in the driest years. The absolute water-table level at CEH36 has continued to decline since 2015 alongside the rest of the network; the improvement is maintained as a relative advantage over the unscraped condition, not as a reversal of the long-term trend. CEH18 and CEH21 were scraped in October 2023 at more seaward locations and have so far shown no detectable benefit in under two years of record; both sites are also subject to coastal-retreat boundary lowering that works against the intervention.

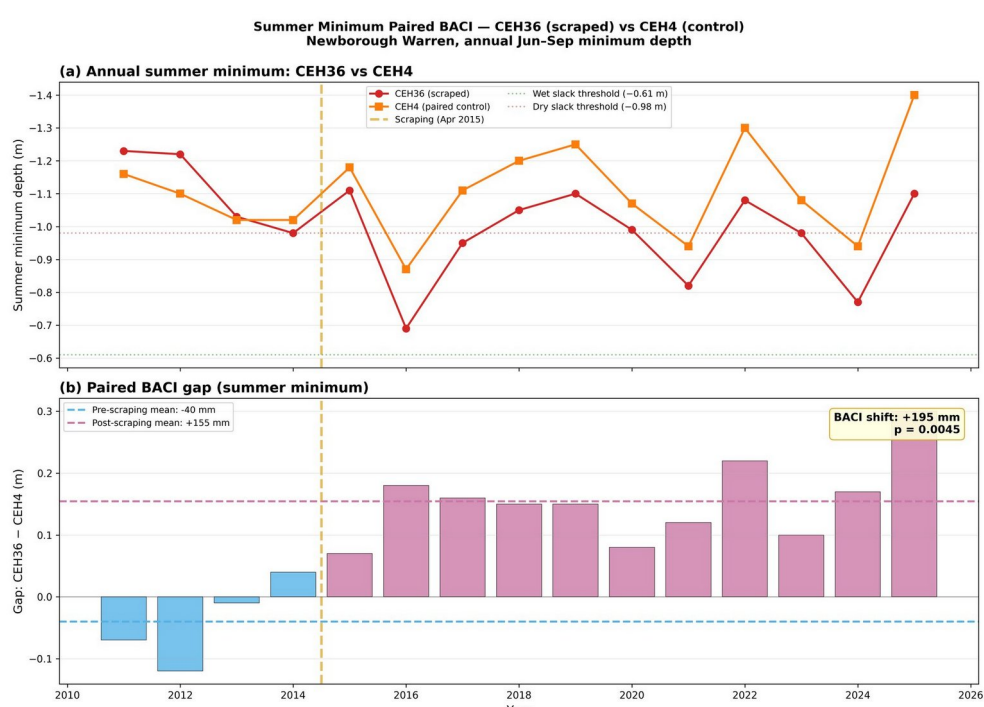


Figure 3. CEH36 (scraped, red) vs control CEH4 (orange). Paired BACI shift: +195 mm ($p = 0.004$). Top: annual summer minima. Bottom: gap between wells before and after scraping.

Tree felling

The December 2017 clearfell of 8.4 ha of Corsican pine produced a statistically significant rise in mean monthly water levels at the core impact well of +120 mm relative to unfelled forest controls ($p < 0.001$). Measured against the ecological yardstick that matters, however, the result is more nuanced. Summer minimum depths — the metric that determines whether dune slack communities can persist — showed no corresponding improvement across any post-felling year. The clearfell raised the annual average water level but produced no detectable benefit at the time of year that determines dune slack survival. The explanation lies in the dual role of the canopy: pine trees intercept rainfall in winter (reducing recharge) but also shade the soil in summer (reducing evaporation). Removing the canopy recovers some winter recharge but increases summer drying by a comparable amount, so the net effect on the summer minimum is near-zero. A further constraint is spatial: the hydraulic divide routes forest groundwater south-westward toward Caernarfon Bay, not eastward toward the Lake Edge and Dune zones where most of the slack ecology sits. Clearfell benefits are largely confined to within the former forest footprint.

A site-wide recharge decline

A decline in recharge efficiency — the proportion of rainfall that reaches the water table — operates across all five zones regardless of land cover or management history. It is consistent with changing rainfall patterns and represents a climate-driven background trend against which all management interventions must be measured. Coastal retreat is a separate and spatially focused driver: as the shoreline erodes, the aquifer loses volume at its seaward edge and the water table adjusts downward across an area extending several hundred metres inland. Because groundwater moves slowly through sand, the declines visible today partly reflect erosion from several years ago. The coastal-retreat signal accounts for approximately 41% of the Coastal Forest zone — exceptional decline — a zone that cannot be restored by scraping or felling alone.

How water levels have changed

Comparing the pre-clearfell five-year window (springs 2013 - 2017) with the most recent comparable window (springs 2019 - 2023), the site-wide spring baseline deepened by an average of 97 mm. Of 59 wells with valid data in both windows, 56 deepened by more than 25 mm; none became shallower by more than 25 mm.

Ecologically, this means that the average dune slack across the site is starting each winter from a position 97 mm worse than it was before the clearfell; a deficit that must be overcome by winter recharge before any flooding benefit reaches the slack vegetation. The largest losses are at the south-western coastal margin; the eastern lake-edge area, where the orchid populations are concentrated, shows the smallest decline.

Looking at how individual zones are moving relative to each other since 2011, the Main Forest (C4) appears to be holding up — but this reflects amplification of recent wet years (2021, 2024) rather than genuine ecological improvement. The forest occupies the topographic and hydraulic high of the aquifer, furthest from any constant-head boundary, so its water table responds more freely to individual wet and dry years than other zones. The Lake Edge (C1) and Coastal Forest (C5) zones are declining relative to the network mean, driven by the retreating shoreline — and C1 is the zone that matters most ecologically.

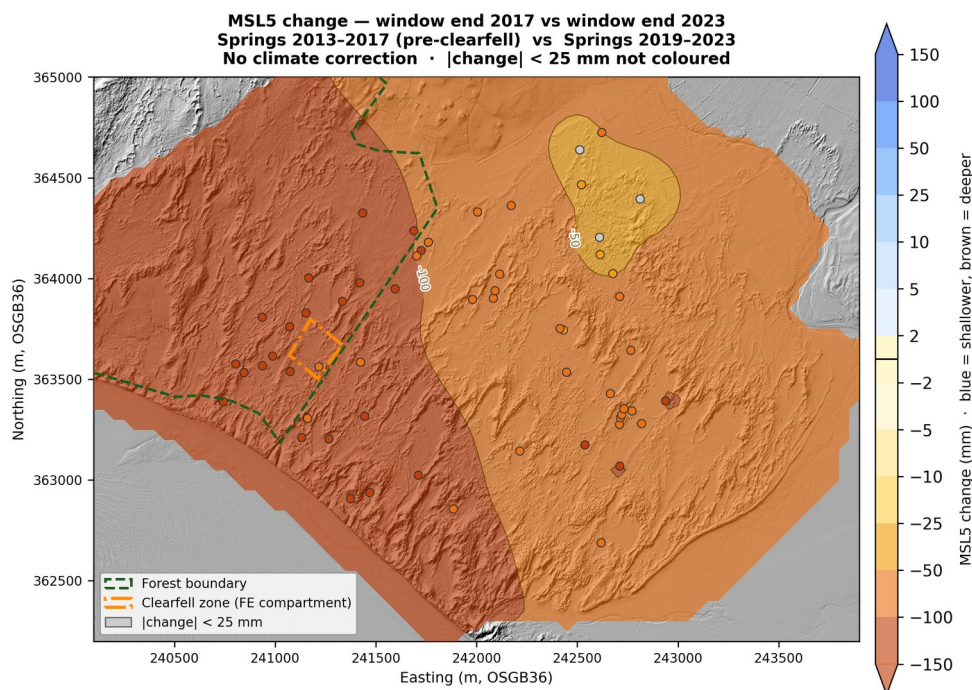


Figure 4. MSL5 change 2017 → 2023: 56 of 59 wells deepened >25 mm; none shallower. Largest losses at south-western coastal margin; clearfell zone (orange) indistinguishable from network trend.

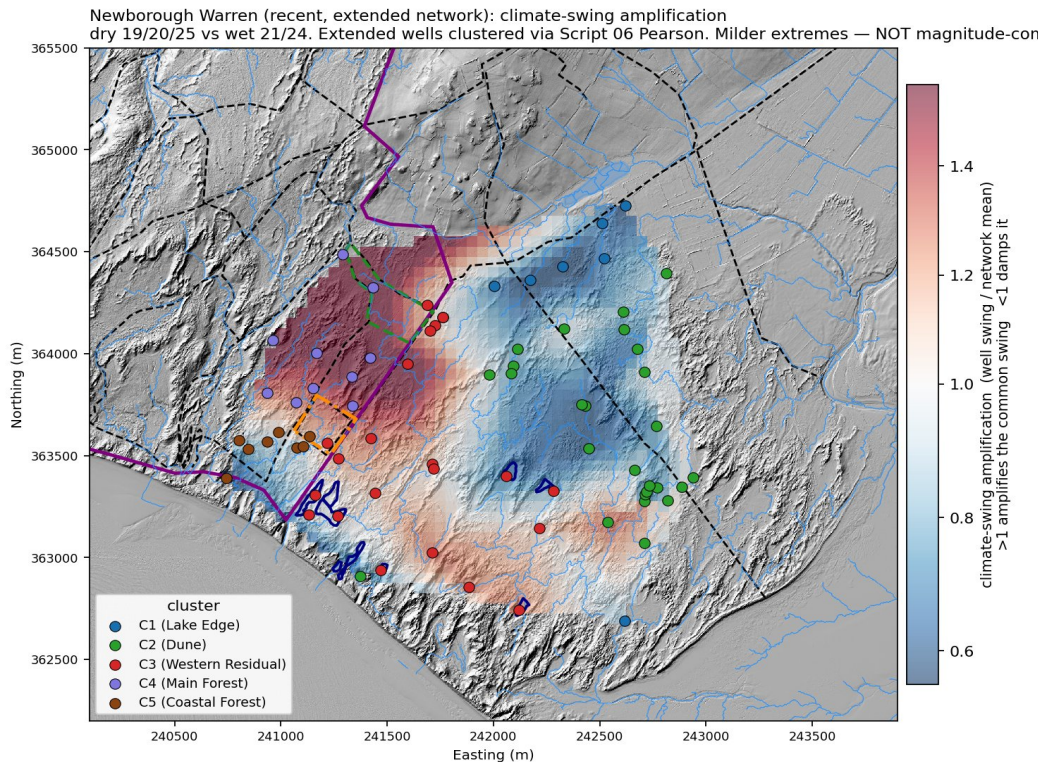


Figure 5. Differential spring movement 2011–2025. Blue = rising relative to network; red = sinking. Forest (C4) uniformly blue from amplification of recent wet springs. Lake Edge (C1) and Coastal Forest (C5) uniformly red from coastal retreat.

The scale of observed change in context

The management interventions studied to date have produced measurable effects at the local scale: the scraping benefit at CEH36 is statistically robust and ecologically meaningful, having shifted that well's summer minimum by 195 mm — enough to move it from regular threshold-breaching to only occasional threshold-breaching. The clearfell produced a detectable improvement in mean monthly water levels within the former forest footprint. These are genuine, quantified achievements.

The challenge is one of scale and rate. The site-wide spring baseline deepened by 97 mm between the 2017 and 2023 comparison windows — a change affecting 56 of 59 monitored wells simultaneously, driven by forces operating at the scale of the whole aquifer. Summer temperatures have trended upward at $+0.014^{\circ}\text{C yr}^{-1}$ since 1931, with a step increase of $+0.94^{\circ}\text{C}$ above baseline since 2013. UKCP18 projections indicate a further summer minimum deepening of 71–134 mm by the 2080s. Added to the 97 mm already lost between the 2017 and 2023 comparison windows, cumulative losses from the pre-clearfell baseline are projected to reach 170–230 mm — comfortably exceeding any management effect observed in this record, and operating simultaneously across the entire network rather than at a single location.

Interactive tools and further information

Three tools are freely available at newbroman.github.io/Newborough_Hydrology. The Scenario Viewer explores how UKCP18 climate projections and management options shift water levels across all five zones. The Flood Forecaster converts a current dipwell reading into a probability of winter flooding, updated through the season with live Met Office rainfall. The Seasonal Extremes Scatter plots every well against the Curreli et al. (2013) ecological viability thresholds. The full technical report, methods supplement and user manuals are available at the same address.